

Statement on Prior Publication and Differences

Present submission: Beyond English-Only GRPO: A Multi-Seed Empirical Study at Sub-3B Scale

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Prior publication disclosed

A preliminary single-seed version of this work (referred to as v1) was self-archived as a preprint on Zenodo on 7 May 2026. The preprint has not undergone peer review.

Citation of v1:

Vu Dang. “Beyond English-Only GRPO: Training Language and Auxiliary Reward as Implicit Regularizers in Sub-3B Math Reasoning,” Zenodo, 7 May 2026. DOI: 10.5281/zenodo.20061328.

The accompanying open-source software repository (github.com/vudang4494/xling-grpo-sub3b, Apache-2.0) is shared between the v1 preprint and the present submission; the repository contains code, configurations, model checkpoints, and evaluation JSONs for both versions.

How the present submission (v2) differs from v1

The submitted manuscript substantially extends the v1 preprint with new experimental content, methodological additions, and revised conclusions. The principal differences are:

1. Multi-seed verification (new experiments). v1 reported single-seed results (seed=42) for three training arms. The present submission expands to three random seeds (seed $\in \{42, 123, 7\}$) per arm. This reveals that v1’s key claims (e.g., +7.5 pp lift on AMC-23, +13.3 pp recovery on AIME-2024 via R_5) fall within seed variance ($\sigma = 11.3$ pp on AMC-23) and require revision. The v2 manuscript reports multi-seed mean $\pm$ standard deviation and bootstrap 95 % confidence intervals for every quantitative claim.
2. Mechanism ablation (new arm). v2 adds Arm A4 (constant-bias reward $R_{\text{const}} = 1.0$, single seed) to test the reward-magnitude mechanism hypothesis proposed in v1. The ablation partially refutes a pure reward-magnitude explanation of the R_5 effect on AIME-2024 maj@8: A4 reverts to base level (33.3 %) while A3 with the actual R_5 reward achieves 37.8 ± 1.9 %. This negative result is new in v2 and is reported honestly as the mechanism remaining an open question.
3. Multilingual evaluation (new benchmark). v2 adds a 10-language MGSM evaluation across all training arms plus the untrained base, comprising 80 new evaluation JSONs (8 conditions $\times$ 10 languages). The result is a negative finding: all training conditions converge to within ± 0.5 pp of the untrained base mean, indicating that training-language choice does not produce cross-lingual transfer at this scale. v1 did not include any MGSM evaluation.
4. Reproducibility infrastructure (new methodology). v2 introduces (a) a SHA256-checksummed manifest covering all 124+ training and evaluation artifacts, (b) bootstrap 95 % confidence intervals for all numerical claims, (c) a 13-section reproducibility audit checklist, and (d) a per-component verification document (VERIFICATION.md). v1 reported numbers without these verification guarantees.

5. Reframed contributions (revised narrative). v1 framed the work as a positive “cross-lingual breakthrough” with auxiliary reward as a regularization mechanism. v2 reframes around three orthogonal findings: (i) positive — A3 best on AIME-2024 maj@8; (ii) methodological — multi-seed instability of vanilla EN GRPO; (iii) negative — no cross-lingual transfer on MGSM. The mechanism hypothesis is presented as an open question rather than a confirmed claim.
6. Compute-cost transparency. v1 contained references to specific monetary compute amounts (e.g., “\$20 single-GPU LoRA budget”). v2 removes all such references for neutrality, describing hardware in physical terms only (“one NVIDIA A100 80 GB GPU”).

Status of v1

The v1 Zenodo preprint remains publicly accessible at its original DOI and is cited within the v2 manuscript as [dang_v1_preprint_2026]. All v1 single-seed numbers are explicitly marked as superseded by the multi-seed results in v2. The author intends to upload the v2 manuscript as a new version of the existing Zenodo DOI upon acceptance, with the v1 version remaining visible for historical reference.

No other prior publication

This manuscript has not been submitted to any peer-reviewed venue, conference, or journal other than IEEE Access. No portions of it have been presented at any conference or workshop. There are no other related publications or preprints by the author covering this work.

Signed:

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